Supplementary document of Fairness-Aware Performance Evaluation for Multi-Party Multi-Objective Optimization

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This appendix contains the details of benchmark and supplementary experiments that omitted in the main paper.

S-I. Parameter Setting in Experiments

A. Parameter setting of constant C

To determine a proper value of the constant C, a preliminary exploration is conducted using NSGA-II. Specifically, NSGA-II is independently executed $R = 10$ times, and the maximum observed loss of each DM is recorded as

$$\widehat{L}_m = \max_{r=1,\dots,R} \max_{\mathbf{x}\in\mathcal{P}^{(r)}_{\text{NSGA-II}}} L_m(\mathbf{x}),$$

where $\mathcal{P}^{(r)}_{\text{NSGA-II}}$ denotes the final population obtained in the r-th run. The constant C is then set as

$$C = \max_{m=1,\dots,M} \widehat{L}_m + \delta,$$

where $\delta > 0$ is a small safety margin.

This choice ensures that C upper-bounds all empirical loss values encountered in the space. NSGA-II is adopted in this preliminary step due to its robustness in approximating diverse Pareto-optimal solutions, making it suitable for estimating an empirical upper bound of loss values.

S-II. Parameter Setting of Penalty Factor λ_m

For each DM m, we project the Pareto points of the other DMs onto the objective subspace of DM m, denoted by $\mathcal{P}_{-m}$. For any $v \in \mathcal{P}_{-m}$, we define the λ_m as

$$\delta_m = \max_{p\in\mathcal{P}_{-m}} \text{IGD}(p, PF_m),$$

$$\lambda_m = \max_{v\in\mathcal{P}_{-m}} \frac{\delta_m - \text{IGD}(v, PF_m)}{\gamma_m(v) - \hat{\gamma}}.$$

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S-III. Computational Cost Analysis

To quantify the computational overhead of the Ψ_{NP}, we conducted a synthetic timing study by varying the number of DMs, M, and the number of objectives per DM, K. For each (M, K) configuration, a solution set containing $N = 200$ points was generated in a two-dimensional decision space. For each DM, 100 reference Pareto-front points were sampled for the party-wise performance evaluation. The objective functions were constructed as linear functions of the decision variables, thereby providing controlled instances with fixed solution-set and reference-set sizes. Ψ_{NP} was independently evaluated 30 times for each configuration. Only the indicator-evaluation time was recorded; data generation, plotting, and file operations were excluded.

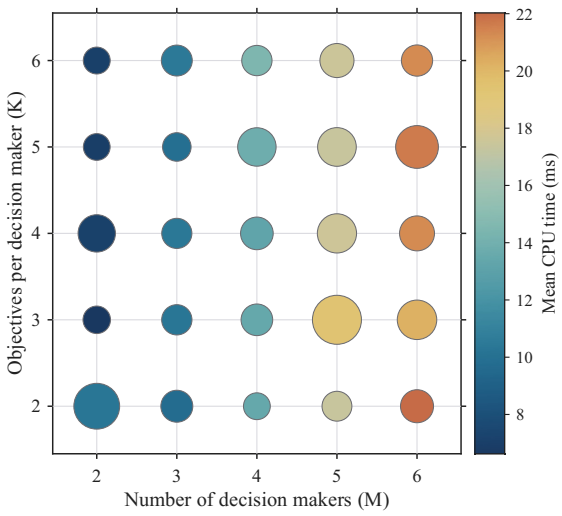


Fig. 1: CPU time of Ψ_{NP} under different numbers of DMs and objectives per DM. Each marker represents a synthetic MPMOP instance with M DMs and K objectives per DM. A solution set with $N = 200$ solutions is evaluated in each case. Marker color denotes the mean CPU time over 30 independent evaluations, whereas marker size is proportional to the corresponding variance.

Fig. 1 presents the resulting CPU-time measurements. The mean evaluation time generally increases with M, since Ψ_{NP} requires party-wise loss evaluation and Nash-product aggregation across all DMs. Nevertheless, the timing variance remains small over the tested configurations, as indicated by the marker

sizes. These results suggest that the proposed indicator incurs stable and manageable computational cost for the multi-party multi-objective problem scales considered in this study.

Let M denote the number of DMs, $N = |P|$ denote the number of solutions in the evaluated set, K_m denote the number of objectives of DM m, and $R_m = |P_m^*|$ denote the number of reference Pareto-optimal solutions used for DM m. For each solution $\mathbf{x} \in P$, computing the concession rate $\gamma_m(x)$ requires evaluating the distance $D_m(\mathbf{x})$ with respect to reference solutions $y \in P_m^*$. Since each $D_m(\mathbf{x})$ involves K_m objective components and the minimum is taken over R_m reference solutions, the cost of computing $\gamma_m(x)$ is $O(R_m K_m)$. The subsequent computation of $\phi_m(x) = \max\{0, \gamma_m(\mathbf{x}) - \hat{\gamma}_m\}$ is $O(1)$ once $\gamma_m(\mathbf{x})$ is available.

Therefore, computing the non-consensus penalty

$$L_m^{\mathrm{pen}}(P) = \sum_{x \in P} \phi_m(x)$$

for DM m requires $O(N R_m K_m)$ time. Across all DMs, the concession-penalty computation requires

$$O\left(N \sum_{m=1}^{M} R_m K_m\right).$$

After the DM-specific losses and penalties are obtained, the Nash-product aggregation requires only $O(M)$ time. Consequently, the total computational complexity of the proposed evaluation framework is

$$O\left(T_L + N \sum_{m=1}^{M} R_m K_m + M\right),$$

where T_L denotes the cost of computing the underlying DM-specific performance losses $L_m(P)$, such as IGD or HV. When the concession rates are precomputed or reused across repeated evaluations under different concession thresholds, the additional penalty and Nash-product aggregation cost reduces to $O(MN + M)$.

S-IV. Search-Induced Asymmetric Sacrifices in MPMOPs

To quantify how frequently mean-based evaluation yields conclusions different from those of the proposed fairness-aware criterion, we compare the best-performing algorithm selected by meanIGD and Ψ_{NP} on each benchmark instance. A result is classified as an agreement when the two indicators select the same algorithm, as a disagreement when they select different algorithms, and as a tie when at least one indicator yields identical best mean values.

Table S-I summarizes the ranking-consistency statistics. Across the 23 benchmark instances for which both meanIGD and Ψ_{NP} are reported, one tie is observed. Among the remaining 22 non-tie comparisons, the two indicators select different best-performing algorithms in 16 cases. In particular, all five MPMOP instances with well-separated individual PFs (MPMOP1–3 and MPMOP7–8) exhibit ranking disagreement. On these problems, OptAll achieves lower meanIGD by improving the average approximation quality across party-wise PFs, whereas Ψ_{NP} favors OptMPNDS because it better avoids unbalanced concessions and non-acceptable solutions.

TABLE S-I: Consistency between the best-performing algorithms selected by meanIGD and Ψ_{NP}.

Problem type	Instances	Agreement	Disagreement	Tie
MPMOP	11	5	5	1
MPDMP	12	1	11	0
Overall	23	6	16	1

In contrast, among MPMOP4–6 and MPMOP9–11, where the party-wise PFs substantially overlap or the conflict between DMs is weaker, the two indicators agree on four of six instances and are tied on one instance. This indicates that mean-based aggregation becomes less problematic when simultaneous improvement for different DMs is feasible. For the MPDMP benchmarks, meanIGD selects OptAll on 11 of 12 instances, whereas Ψ_{NP} generally favors the multiparty optimizers. This systematic discrepancy arises because OptAll is optimized in the concatenated objective space and therefore naturally performs well on the geometric mean-based metric, while Ψ_{NP} additionally accounts for party-wise balance and mutual acceptability.

To clarify under what conditions meanIGD may conceal uneven compromises among DMs, this subsection examines the search behaviors of representative MPMOP optimizers. Asymmetric sacrifices are jointly determined by the geometry of the party-wise Pareto sets and the selection mechanisms used during optimization. We therefore categorize existing MPMOP algorithms according to their principal search strategies, including aggregate-objective optimization, multiparty non-dominated sorting, party-by-party decomposition, and archive- or clone-based search. For each category, we discuss the circumstances under which its search process may lead to unfair outcomes, and we use two representative cases to illustrate the analysis.

For aggregate-objective methods such as OptAll, all party-wise objectives are treated as a single objective vector. Consequently, solutions favoring different DMs can remain non-dominated in the concatenated objective space, even when a solution is substantially inferior from the perspective of one DM. This behavior is particularly likely when the party-wise Pareto sets have limited overlap or are disjoint, since improving the approximation quality for one DM may require moving the population away from the preferred region of another DM. Thus, a low meanIGD may be obtained by combining a small loss for one DM with a much larger loss for another.

Multiparty non-dominated sorting methods, including OptMPNDS [1], OptMPNDS2 [2], and OptMPNDS3 [3], reduce this risk by explicitly considering the Pareto status of every DM. However, these methods mainly exploit ordinal rank information. Hence, solutions with substantially different sacrifice magnitudes may remain indistinguishable when their party-wise rank vectors are mutually non-dominated. For ex-

ample, two solutions with rank vectors $(1, 2)$ and $(6, 1)$ can receive comparable multiparty priority, although one solution is much farther from the Pareto set of one DM. Therefore, multiparty sorting improves party-wise Pareto consistency but does not guarantee balanced approximation losses.

For party-by-party or decomposition-based search as MOEA/D-MP [4], asymmetric sacrifices may arise when the DMs have unequal approximation difficulty. When one DM has a smooth and easily accessible Pareto structure, whereas another has a disconnected, non-convex, or multimodal Pareto structure, the population may converge more rapidly toward the former. Although the objectives of all DMs are considered sequentially, later updates may not fully compensate for the slower progress on the more difficult party.

Archive- or clone-based methods like MPNNIA [5] may further amplify an initial party-wise imbalance through unequal survival or reproduction opportunities. In particular, if solutions preferred by one DM consistently receive better ranks, lower local density, or more cloning resources, that region can progressively dominate the population. Such mechanisms may preserve diversity within the favored region while reducing the search effort devoted to the other DM.

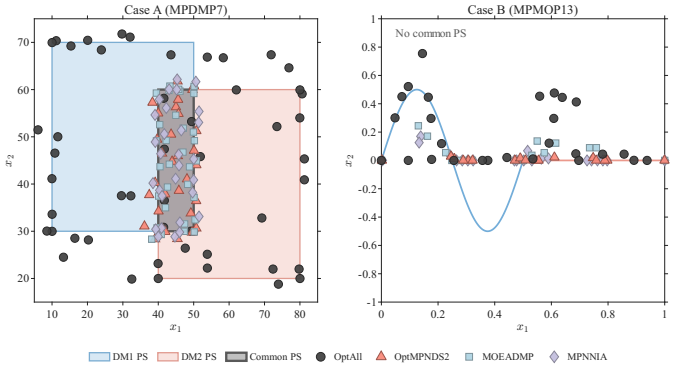


Fig. 2: Behavioral comparison of representative MPMOP optimizers under two distinct party-wise geometries. (a) Case A (MPDMP7): the two DMs have geometrically similar Pareto sets with a nonempty overlap region, yielding a common PS where both parties can be simultaneously satisfied. (b) Case B (MPMOP13): the two DMs have geometrically dissimilar Pareto sets with no common PS; in particular, the multimodal Pareto structure of DM1 is more difficult to approximate than the linear Pareto set of DM2.

Fig. 2 illustrates the above mechanisms under two representative party-wise geometries, and Table S-II gives the specific IGD score on both DMs. In Case A, the two PSs are geometrically similar and have a nonempty overlap region. The multiparty methods, including OptMPNDS2, MOEA/D-MP, and MPNNIA, place most of their final solutions within or close to the common PS. Therefore, both DMs can be improved simultaneously and substantial asymmetric sacrifices are unlikely. In contrast, OptAll retains a broader solution distribution over the two individual PS regions. Although these solutions may remain non-dominated after objective concatenation, many lie outside the common PS and are inferior from the viewpoint of at least one DM.

TABLE S-II: Party-wise IGD values of the representative final populations on Case A (MPDMP7) and Case B (MPMOP13). IGD_1 and IGD_2 denote the inverted generational distance with respect to the reference Pareto front of DM1 and DM2, respectively. Mean IGD is the arithmetic average of the two party-wise IGD values.

	Case A (MPDMP7)			Case B (MPMOP13)		
Algorithm	IGD_1	IGD_2	Mean	IGD_1	IGD_2	Mean
OptAll	5.747	5.437	5.592	0.0197	0.0071	0.0134
OptMPNDS2	15.852	15.849	15.851	0.0540	0.0545	0.0542
MOEA/D-MP	15.961	15.558	15.760	0.0381	0.0351	0.0366
MPNNIA	16.062	15.623	15.842	0.0396	0.0363	0.0380

Case B has no common PS and exhibits both geometric dissimilarity and unequal approximation difficulty. The linear PS of DM2 is relatively easy to approach, whereas the multimodal PS of DM1 contains separated preferred regions and is more difficult to approximate. Consequently, a solution set may achieve a favorable average approximation level while still leaving DM1 farther from its PS. OptAll produces a broad distribution including solutions close to both DMs' PS, while the multiparty methods focus more strongly on the DM2's PS. These observations show that multiparty mechanisms do not ensure equal sacrifice magnitudes when the party-wise PSs are separated and their approximation difficulties differ.

Overall, mean-based evaluation is most vulnerable to masking unequal compromises when the party-wise PSs have limited overlap or no intersection and when the approximation difficulty differs substantially across DMs. In these settings, the mean loss can remain small even when the loss is concentrated on one DM. The proposed Ψ_{NP} complements meanIGD by explicitly accounting for both party-wise losses and mutual acceptability, thereby distinguishing balanced compromises from solutions that achieve a favorable average performance at the expense of a particular DM.

S-V. Validity of Nash-Product Evaluation over Non-convex Utility Sets

Classical Nash bargaining usually assumes convexity of the feasible utility set when the Nash product is maximized to obtain a bargaining solution with existence and uniqueness guarantees. In the proposed framework, however, the Nash product is used only to evaluate obtained solution sets rather than to select a unique bargaining outcome. Therefore, the attainable utility set is not required to be convex.

Lemma 1. *Let $\mathcal{U} \subseteq \mathbb{R}{+}{+}^M$ be an arbitrary attainable utility set, possibly non-convex. For any $\mathbf{u} = (u_1, \ldots, u_M) \in \mathcal{U}$, the Nash-product indicator*

$$\Psi_{\mathrm{NP}}(\mathbf{u}) = \prod_{m=1}^{M} u_m \tag{1}$$

is well defined. Moreover, if $\mathbf{u}, \mathbf{v} \in \mathcal{U}$ *satisfy* $u_m \geq v_m$ *for all* m *and* $u_j > v_j$ *for at least one* j*, then*

$$\Psi_{\mathrm{NP}}(\mathbf{u}) > \Psi_{\mathrm{NP}}(\mathbf{v}). \tag{2}$$

Proof. Since $\mathcal{U} \subseteq \mathbb{R}{+}{+}^M$, all utilities are positive; hence, $\Psi_{\mathrm{NP}}(\mathbf{u})$ is well defined for every attainable utility vector. Further, the product of positive numbers is strictly increasing in each component. Thus, any componentwise utility improvement strictly increases Ψ_{NP}. These arguments are pointwise and do not depend on the convexity of $\mathcal{U}$. □

Remark 1. *If* Ψ_{NP} *were maximized over a non-convex utility set, multiple maximizers could exist. This issue is irrelevant here because* Ψ_{NP} *is used as an evaluation indicator for comparing obtained solution sets, rather than as a rule for selecting a unique bargaining solution.*

S-VI. Convergence Behavior on Representative MPDMPs

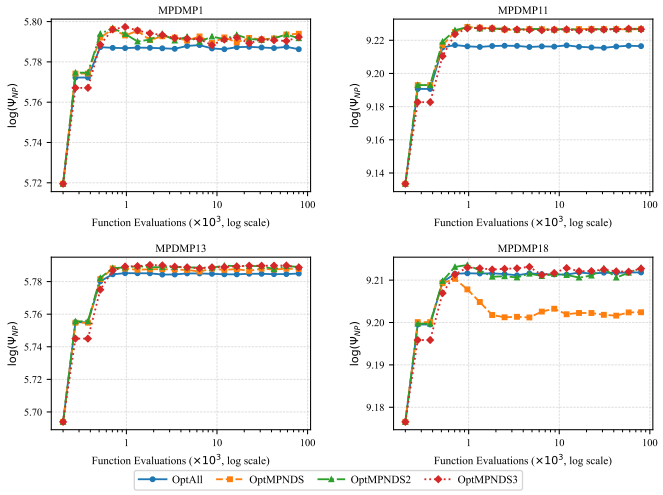


Fig. 3: Convergence curves of the compared algorithms on representative MPDMPs. The horizontal axis denotes the number of function evaluations on a logarithmic scale.

To examine the search dynamics of the compared algorithms, Fig. 3 presents the convergence curves on four representative MPDMP instances. MPDMP1 and MPDMP11 are selected from the original benchmark suite, whereas MPDMP13 and MPDMP18 are selected from the extended no-common-PS suite. The logarithmic form is adopted to avoid numerical scaling issues and to facilitate visual comparison. All algorithms show rapid improvement during the early stage, indicating that the search processes quickly move away from the initial random populations toward higher-quality regions. On MPDMP1, MPDMP11, and MPDMP13, the curves gradually stabilize after the initial increase, suggesting that the algorithms reach relatively steady search states. The multiparty methods generally achieve higher final plateaus than OptAll, while the differences among OptMPNDS, OptMPNDS2, and OptMPNDS3 are comparatively small on MPDMP1 and MPDMP11. The convergence behaviors become more distinguishable on MPDMP18. Although OptMPNDS initially attains a competitive value, its curve decreases during the subsequent search stage, indicating that the population quality is not fully preserved. In contrast, OptMPNDS2 and OptMPNDS3 maintain relatively stable late-stage trajectories, while OptMPNDS3 reaches the highest final value. These results indicate that the compared algorithms differ not only in their final performance but also in their ability to maintain population quality during evolution.

S-VII. Limitations and Sensitivity Analyses

To isolate indicator behaviour from optimizer-dependent effects, we conduct a metric-level case study on MPDMP13 with two DMs. The decision space is $\Omega = [0, 100]^2$, where the two DMs minimize distances to line segments located at $y = 40$ and $y = 60$, respectively. No optimizer is executed. Instead, two fixed candidate sets, P_{bias} and P_{fair}, each containing 100 decision vectors, are evaluated under identical settings. P_{bias} is constructed to favor DM1: 85 solutions are sampled from DM1's Pareto set, and the remaining 15 solutions are generated near this set with a vertical perturbation bounded by 3. In contrast, P_{fair} contains 100 solutions sampled from line $y = 50$.

a) Sensitivity to objective dimensionality: Fig. 4 reports the results obtained by increasing the number of objectives per DM from 2 to 20. Higher-dimensional objective vectors are constructed through deterministic positive affine replications of the original objectives. This construction is intended as a controlled stress test of indicator behaviour rather than as a replacement for a dedicated high-dimensional benchmark suite.

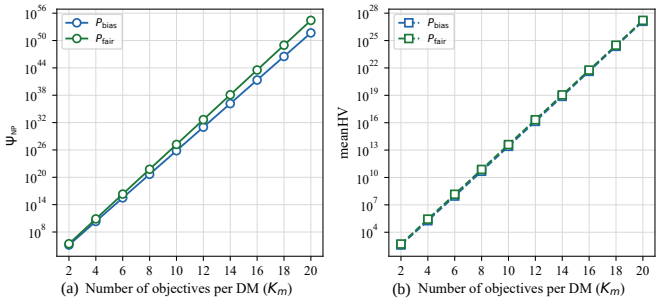


Fig. 4: Sensitivity of Ψ_{NP} and meanHV to the number of objectives per DM. P_{bias} predominantly favors DM1, whereas P_{fair} provides balanced coverage of the two party-wise Pareto sets.

As shown in Fig. 4(a), Ψ_{NP} consistently assigns a higher score to P_{fair} than to P_{bias} across all tested dimensions. This result reflects the ability of the product aggregation to penalize the imbalance induced by the DM1-favoring set. In contrast, the meanHV values of the two sets are nearly indistinguishable in Fig. 4(b), indicating that averaging party-wise HV values may conceal the unequal coverage of the two DMs. The absolute values of both indicators increase rapidly as K_m

grows. This behaviour reflects the known numerical scaling difficulty of raw HV values in higher-dimensional spaces. Therefore, although the product-based aggregation remains discriminative in this controlled example, implementations should be used cautiously when the number of objectives is large.

b) Sensitivity to reference-point selection: To investigate reference-point sensitivity, we fix $K_m = 8$ and vary the expansion factor as

$$\delta \in \{0.05, 0.10, 0.20, 0.50, 1.00\}. \tag{3}$$

For DM m, the reference point is defined as

$$r_m(\delta) = z_m^{\text{nadir}} + \delta \left(z_m^{\text{nadir}} - z_m^{\text{ideal}} \right), \tag{4}$$

where z_m^{ideal} and z_m^{nadir} are estimated from the union of PF samples, P_{bias}, and P_{fair}.

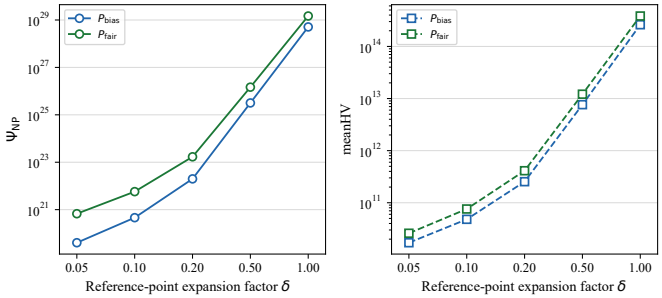


Fig. 5: Sensitivity of HV-based indicators to the reference-point expansion factor δ for the controlled MPDMP13 case with $K_m = 8$.

Fig. 5 shows that both meanHV and Ψ_{NP} increase substantially as δ becomes larger. Hence, their absolute values are sensitive to the location of the reference point. Nevertheless, Ψ_{NP} consistently ranks P_{fair} above P_{bias} under all tested values of δ, whereas the separation under meanHV remains comparatively small. This ranking stability should not be interpreted as reference-point invariance. Rather, it indicates that the Ψ_{NP} retains its preference for balanced party-wise coverage under the tested reference-point range. In practical applications, all compared solution sets should be evaluated using the same reference-point construction, and the reference point should be selected based on a justified estimate of the attainable objective range.

S-VIII. Comparison With Representative Aggregation Rules

Table S-III compares several representative aggregation rules with the four axioms. The purpose is not to claim that the Nash product is uniquely characterized by these axioms, but to show that commonly used alternatives do not satisfy all of them simultaneously.

The arithmetic mean is Pareto-monotone and symmetric, but it assigns identical scores to balanced and unbalanced utility vectors with the same total utility and therefore violates Axiom 3. The minimum-utility rule violates strict Pareto Monotonicity because improving a non-worst-off DM may leave the aggregate unchanged. Jain's index and $1-\text{Gini}$ favor balanced utility distributions, but they are not strictly Pareto-monotone: for example, moving from $(1,1)$ to $(2,1)$ improves one DM without harming the other, while both fairness values decrease. The Nash product satisfies Axioms 1–3 for positive utilities, but it does not itself penalize violations of the jointly acceptable region. The concession-based penalty in Ψ_{NP} is therefore required to enforce Axiom 4.

TABLE S-III: Comparison of representative aggregation rules with the proposed axioms.

Aggregation rule	A1	A2	A3	A4
Arithmetic mean	✓	✓	×	×
Minimum utility	×	✓	×	×
Jain's index / $1-\text{Gini}$	×	✓	✓	×
Nash product	✓	✓	✓	×
Ψ_{NP}	✓	✓	✓	✓

S-IX. MPMOP12-17 Benchmark Problems

This section presents detailed descriptions of the MPMOP12–MPMOP17 benchmark problems, which are specifically constructed to ensure that the PSs of different DMs are mutually disjoint, i.e., no solution is Pareto-optimal for all DMs simultaneously. This characteristic distinguishes MPMOP12–MPMOP17 from MPMOP1–11, in which common Pareto-optimal solutions across DMs may exist. The detailed configurations of these benchmark problems are summarized in Table S-VI.

To evaluate algorithmic behavior under varying levels of compromise among DMs, multiple experimental scenarios are further defined using different concession-rate thresholds. The same set of concession-rate thresholds is applied consistently to both categories of benchmark problems, enabling a unified and fair comparison across problem types. The considered threshold settings are reported in Table S-IV.

S-X. MPDMP13-18 Benchmark Problems

The MPDMP13–MPDMP18 benchmark suite extends the MPDMP framework by considering more complex geometric configurations with multiple decision makers (DMs) and heterogeneous numbers of objectives. These problems are constructed using different Pareto set (PS) geometries, including parallel line segments, triangles, squares, and pentagons, which induce diverse trade-off structures among DMs. As a result, the suite provides controlled yet nontrivial test scenarios for evaluating algorithm performance in multi-DM optimization settings.

For all problems, the decision vector is defined on a two-dimensional continuous space,$\mathbf{x} \in [0, 100]^2$. For each DM $m \in \{1, \ldots, M\}$, let

$$F_m(\mathbf{x}) = \left(f_{m,1}(\mathbf{x}), \ldots, f_{m,K_m}(\mathbf{x}) \right)$$

denote its objective vector. Each objective function is defined as the Euclidean distance from the decision vector $\mathbf{x}$ to

a reference point associated with the underlying geometric structure.

$$f_{m,i}(\mathbf{x}) = \sqrt{(x_1 - p_{m,i,1})^2 + (x_2 - p_{m,i,2})^2}, \quad i = 1, \ldots, K_m, \tag{5}$$

where $\mathbf{p}_{m,i} = [p_{m,i,1}, p_{m,i,2}]^\mathsf{T}$ denotes the i-th reference point defining the geometric structure for DM m, and K_m is the number of objectives associated with DM m. The reference points for all benchmark problems are summarized in Table S-V.

TABLE S-IV: Concession Rate Threshold Configurations

Case	**Description**	**2 DMs**		**3 DMs**		
		γ_1	γ_2	γ_1	γ_2	γ_3
1	No concession	0.0	0.0	0.0	0.0	0.0
2	Partial concession	0.3	0.0	0.3	0.0	0.0
3	Equal partial	0.3	0.3	0.3	0.3	0.3
4	Asymmetric	0.3	0.7	0.3	0.1	0.6
5	Equal moderate	0.5	0.5	0.5	0.5	0.5
6	Equal high	0.7	0.7	0.7	0.7	0.7

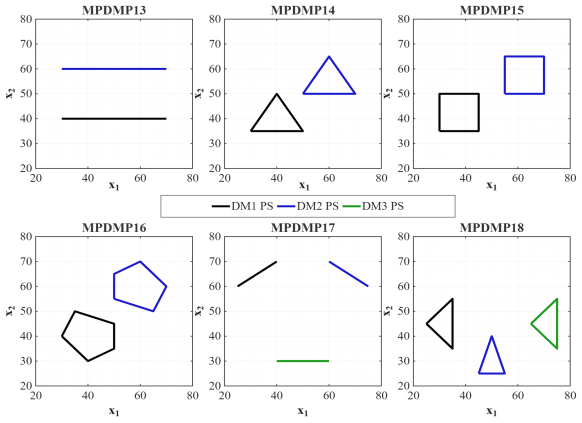


Fig. 6: MPDMPs with no common Pareto optimal solutions.

TABLE S-V: Reference points for MPDMP13–MPDMP18

Problem	DM	Objective number	PS	$p_{i,1}$	$p_{i,2}$
MPDMP13	1	2	Line segment	30	40
				70	40
	2	2	Line segment	30	60
				70	60
MPDMP14	1	3	Triangle	30	35
				50	35
				40	50
	2	3	Triangle	50	50
				70	50
				60	65
MPDMP15	1	4	Square	30	35
				45	35
				45	50
				30	50
	2	4	Square	55	50
				70	50
				70	65
				55	65
MPDMP16	1	5	Pentagon	30	40
				40	30
				50	35
				50	45
				35	50
	2	5	Pentagon	50	55
				65	50
				70	60
				60	70
				50	65
MPDMP17	1	2	Line segment	25	60
				40	70
	2	2	Line segment	60	70
				75	60
	3	2	Line segment	40	30
				60	30
MPDMP18	1	3	Triangle	25	45
				35	35
				35	55
	2	3	Triangle	45	25
				55	25
				50	40
	3	3	Triangle	65	45
				75	35
				75	55

TABLE S-VI: Mathematical Definitions of Benchmark Problems MPMOP12–MPMOP17

Problem	DM	Objective Functions	Variable Bounds	Pareto Set (PS)
MPMOP12	DM_1	$F_1 = (f_{11}(\mathbf{x}), f_{12}(\mathbf{x}))$, $g_1(\mathbf{x}) = 1 + \sum_{i=2}^{D}(x_i - 0.5)^2$, $f_{11}(\mathbf{x}) = g_1(\mathbf{x}) \cdot \frac{2}{x_1}, \quad f_{12}(\mathbf{x}) = g_1(\mathbf{x}) \cdot \frac{x_1}{2}$	$x_1 \in [1,2]$, $x_{i\geq 2} \in [0,1]$	$\{\mathbf{x} \mid x_1 \in [1,2],\ x_i = 0.5\ (i \geq 2)\}$
	DM_2	$F_2 = (f_{21}(\mathbf{x}), f_{22}(\mathbf{x}))$, $g_2(\mathbf{x}) = 1 + \sum_{i=2}^{D}\left(x_i - \frac{1}{1+e^{-5(x_1-3.5)}}\right)^2$, $f_{21}(\mathbf{x}) = g_2(\mathbf{x}) \cdot \frac{3}{x_1}, \quad f_{22}(\mathbf{x}) = g_2(\mathbf{x}) \cdot \frac{x_1}{3}$	$x_1 \in [3,4]$, $x_{i\geq 2} \in [0,1]$	$\{\mathbf{x} \mid x_1 \in [3,4],\ x_i = \frac{1}{1+e^{-5(x_1-3.5)}}\ (i \geq 2)\}$
MPMOP13	DM_1	$F_1 = (f_{11}(\mathbf{x}), f_{12}(\mathbf{x}))$, $g_1(\mathbf{x}) = 1 + \sum_{i=2}^{D}\left(x_i - \frac{\sin(4\pi x_1)}{2}\right)^2 + 100\max(0, x_1 - 0.5)^2$, $f_{11}(\mathbf{x}) = g_1(\mathbf{x})\big(x_1 + 0.1\sin(3\pi x_1)\big)$, $f_{12}(\mathbf{x}) = g_1(\mathbf{x})\big(1 - x_1 + 0.1\sin(3\pi x_1)\big)^{4.25}$	$x_i \in [0,1]$	$\{\mathbf{x} \mid x_1 \in [0,0.5),\ x_i = \frac{\sin(4\pi x_1)}{2}\ (i \geq 2)\}$
	DM_2	$F_2 = (f_{21}(\mathbf{x}), f_{22}(\mathbf{x}))$, $g_2(\mathbf{x}) = 1 + \sum_{i=2}^{D} x_i^2 + 100\max(0, 0.5 - x_1)^2$, $f_{21}(\mathbf{x}) = g_2(\mathbf{x})\big(x_1 + 0.1\sin(3\pi x_1)\big)$, $f_{22}(\mathbf{x}) = g_2(\mathbf{x})\big(1 - x_1 + 0.1\sin(3\pi x_1)\big)^{4.25}$	$x_i \in [0,1]$	$\{\mathbf{x} \mid x_1 \in [0.5,1],\ x_i = 0\ (i \geq 2)\}$
MPMOP14	DM_1	$F_1 = (f_{11}(\mathbf{x}), f_{12}(\mathbf{x}), f_{13}(\mathbf{x}))$, $g_1(\mathbf{x}) = 1 + \sum_{i=3}^{D}\left(x_i - \frac{\sin(2\pi(x_1+x_2)+\pi/4)}{2}\right)^2$, $f_{11}(\mathbf{x}) = g_1(\mathbf{x})\sin^2\left(\frac{\pi}{2}x_1\right)$, $f_{12}(\mathbf{x}) = g_1(\mathbf{x})\sin^2\left(\frac{\pi}{2}x_2\right)\cos^2\left(\frac{\pi}{2}x_1\right)$, $f_{13}(\mathbf{x}) = g_1(\mathbf{x})\cos^2\left(\frac{\pi}{2}x_2\right)\cos^2\left(\frac{\pi}{2}x_1\right)$	$x_i \in [0,1]$	$\{\mathbf{x} \mid x_1 \in [0,0.5],\ x_2 \in [0,1],\ x_i = \frac{\sin(2\pi(x_1+x_2)+\pi/4)}{2}\ (i \geq 3)\}$
	DM_2	$F_2 = (f_{21}(\mathbf{x}), f_{22}(\mathbf{x}), f_{23}(\mathbf{x}))$, $g_2(\mathbf{x}) = 1 + \sum_{i=3}^{D}\left(x_i - \sin\left(2\pi(x_1+x_2)+\frac{\pi}{2}\right)\right)^2$, $f_{21}(\mathbf{x}) = g_2(\mathbf{x})\sin^2\left(\frac{\pi}{2}x_1\right)$, $f_{22}(\mathbf{x}) = g_2(\mathbf{x})\sin^2\left(\frac{\pi}{2}x_2\right)\cos^2\left(\frac{\pi}{2}x_1\right)$, $f_{23}(\mathbf{x}) = g_2(\mathbf{x})\cos^2\left(\frac{\pi}{2}x_2\right)\cos^2\left(\frac{\pi}{2}x_1\right)$	$x_i \in [0,1]$	$\{\mathbf{x} \mid x_1 \in (0.5,1],\ x_2 \in [0,1],\ x_i = \sin\left(2\pi(x_1+x_2)+\frac{\pi}{2}\right)\ (i \geq 3)\}$

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TABLE S-VI: Mathematical Definitions of Benchmark Problems MPMOP12–MPMOP17 (Continued)

Prob.	DM	Objective Functions	Variable Bounds	Pareto Set (PS)
MPMOP15		$F_1 = (f_{11}(\mathbf{x}), f_{12}(\mathbf{x}))$,		
	DM_1	$g_1(\mathbf{x}) = 1 + \sum_{i=2}^{D}(x_i - 0.5)^2$, $f_{11}(\mathbf{x}) = g_1(\mathbf{x})\frac{2}{x_1}, \quad f_{12}(\mathbf{x}) = g_1(\mathbf{x})\frac{x_1}{2}$	$x_1 \in [1,2]$, $x_{i\geq 2} \in [0,1]$	$\{\mathbf{x} \mid x_1 \in [1,2],\ x_i = 0.5\ (i \geq 2)\}$
		$F_2 = (f_{21}(\mathbf{x}), f_{22}(\mathbf{x}))$,		
	DM_2	$g_2(\mathbf{x}) = 1 + \sum_{i=2}^{D}\left(x_i - \frac{1}{1+e^{-5(x_1-2.5)}}\right)^2$, $f_{21}(\mathbf{x}) = g_2(\mathbf{x})\frac{3}{x_1}, \quad f_{22}(\mathbf{x}) = g_2(\mathbf{x})\frac{x_1}{3}$	$x_1 \in [2,3]$, $x_{i\geq 2} \in [0,1]$	$\{\mathbf{x} \mid x_1 \in [2,3],\ x_i = \frac{1}{1+e^{-5(x_1-2.5)}}\ (i \geq 2)\}$
		$F_3 = (f_{31}(\mathbf{x}), f_{32}(\mathbf{x}))$,		
	DM_3	$g_3(\mathbf{x}) = 1 + \sum_{i=2}^{D}(x_i - 0.5)^2$, $f_{31}(\mathbf{x}) = g_3(\mathbf{x})\frac{4}{x_1}, \quad f_{32}(\mathbf{x}) = g_3(\mathbf{x})\frac{x_1}{4}$	$x_1 \in [3,4]$, $x_{i\geq 2} \in [0,1]$	$\{\mathbf{x} \mid x_1 \in [3,4],\ x_i = 0.5\ (i \geq 2)\}$
MPMOP16		$F_1 = (f_{11}(\mathbf{x}), f_{12}(\mathbf{x}))$,		
	DM_1	$g_1(\mathbf{x}) = 1 + \sum_{i=2}^{D}\left(x_i - \frac{\sin(4\pi x_1)}{2}\right)^2 + 100\max(0, x_1 - 1/3)^2$, $f_{11}(\mathbf{x}) = g_1(\mathbf{x})\big(x_1 + 0.1\sin(3\pi x_1)\big)$, $f_{12}(\mathbf{x}) = g_1(\mathbf{x})\big(1 - x_1 + 0.1\sin(3\pi x_1)\big)^{2.5}$	$x_i \in [0,1]$	$\{\mathbf{x} \mid x_1 \in [0,1/3),\ x_i = \frac{\sin(4\pi x_1)}{2}\ (i \geq 2)\}$
		$F_2 = (f_{21}(\mathbf{x}), f_{22}(\mathbf{x}))$,		
	DM_2	$g_2(\mathbf{x}) = 1 + \sum_{i=2}^{D} x_i^2 + 100\left(\max(0, 1/3 - x_1)^2 + \max(0, x_1 - 2/3)^2\right)$, $f_{21}(\mathbf{x}) = g_2(\mathbf{x})\big(x_1 + 0.1\sin(3\pi x_1)\big)$, $f_{22}(\mathbf{x}) = g_2(\mathbf{x})\big(1 - x_1 + 0.1\sin(3\pi x_1)\big)^{3.0}$	$x_i \in [0,1]$	$\{\mathbf{x} \mid x_1 \in [1/3, 2/3),\ x_i = 0\ (i \geq 2)\}$
		$F_3 = (f_{31}(\mathbf{x}), f_{32}(\mathbf{x}))$,		
	DM_3	$g_3(\mathbf{x}) = 1 + \sum_{i=2}^{D}\left(x_i + \frac{\sin(4\pi x_1)}{2}\right)^2 + 100\max(0, 2/3 - x_1)^2$, $f_{31}(\mathbf{x}) = g_3(\mathbf{x})\big(x_1 + 0.1\sin(3\pi x_1)\big)$, $f_{32}(\mathbf{x}) = g_3(\mathbf{x})\big(1 - x_1 + 0.1\sin(3\pi x_1)\big)^{3.5}$	$x_i \in [0,1]$	$\{\mathbf{x} \mid x_1 \in [2/3, 1],\ x_i = -\frac{\sin(4\pi x_1)}{2}\ (i \geq 2)\}$

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TABLE S-VI: Mathematical Definitions of Benchmark Problems MPMOP12–MPMOP17 (Continued)

Prob.	DM	Objective Functions	Variable Bounds	Pareto Set (PS)
MPMOP17	DM$_1$	$F_1 = (f_{11}(\mathbf{x}), f_{12}(\mathbf{x}), f_{13}(\mathbf{x}))$, $g_1(\mathbf{x}) = 1 + \sum_{i=3}^{D}(x_i - 0.5x_1)^2$, $y_1 = \frac{\pi}{6} + \frac{\pi}{3}x_1, \quad y_2 = \frac{\pi}{2}x_2$, $f_{11}(\mathbf{x}) = g_1(\mathbf{x})\sin(y_1)$, $f_{12}(\mathbf{x}) = g_1(\mathbf{x})\sin(y_2)\cos(y_1)$, $f_{13}(\mathbf{x}) = g_1(\mathbf{x})\cos(y_2)\cos(y_1)$	$x_i \in [0,1]$	$\{\mathbf{x} \mid x_1 < 0.33,\ x_2 \in [0,1],\ x_i = 0.5x_1\ (i \geq 3)\}$
	DM$_2$	$F_2 = (f_{21}(\mathbf{x}), f_{22}(\mathbf{x}), f_{23}(\mathbf{x}))$, $g_2(\mathbf{x}) = 1 + \sum_{i=3}^{D}x_i^2$, $y_1 = \frac{\pi}{2}x_1, \quad y_2 = \frac{\pi}{2}x_2$, $f_{21}(\mathbf{x}) = g_2(\mathbf{x})\sin(y_1)$, $f_{22}(\mathbf{x}) = g_2(\mathbf{x})\sin(y_2)\cos(y_1)$, $f_{23}(\mathbf{x}) = g_2(\mathbf{x})\cos(y_2)\cos(y_1)$	$x_i \in [0,1]$	$\{\mathbf{x} \mid 0.33 \leq x_1 < 0.66,\ x_2 \in [0,1],\ x_i = 0\ (i \geq 3)\}$
	DM$_3$	$F_3 = (f_{31}(\mathbf{x}), f_{32}(\mathbf{x}), f_{33}(\mathbf{x}))$, $g_3(\mathbf{x}) = 1 + \sum_{i=3}^{D}(x_i - 0.5x_1)^2$, $y_1 = \frac{\pi}{6} + \frac{\pi}{3}x_1, \quad y_2 = \frac{\pi}{2}x_2$, $f_{31}(\mathbf{x}) = g_3(\mathbf{x})\sin(y_1)$, $f_{32}(\mathbf{x}) = g_3(\mathbf{x})\sin(y_2)\cos(y_1)$, $f_{33}(\mathbf{x}) = g_3(\mathbf{x})\cos(y_2)\cos(y_1)$	$x_i \in [0,1]$	$\{\mathbf{x} \mid x_1 \geq 0.66,\ x_2 \in [0,1],\ x_i = 0.5x_1\ (i \geq 3)\}$

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